IMPLEMENTATION OF AUKF SOC BASED ESTIMATION FOR LITHIUM ION BATTERY IN SIL CONFIGURATION

A SUMMER INTERNSHIP PROJECT REPORT

submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

This project presents a comprehensive methodology for accurate State of Charge (SOC) estimation of lithium-ion batteries using an Adaptive Square Root Unscented Kalman Filter (A-SRUKF), validated within a Software in the loop(SIL) configuration. The workflow begins with modeling the battery using a 2-RC equivalent circuit model (ECM), where parameters such as resistances and capacitances are extracted and tuned region-wise using the Extended Kalman Filter (EKF) based on HPPC test data. These temperature- and SOC-dependent parameters are organized into 2D lookup tables to dynamically reflect the nonlinear behavior of the battery under varying operating conditions.

The second phase involves the development and implementation of an A-SRUKF-based SOC estimator in MATLAB, formulated within a MATLAB Function Block for direct integration with Simulink. This estimator accounts for non-linearities in the ECM, including time-varying internal resistances, polarization voltages, and open circuit voltage (OCV) dynamics. The estimator block receives real-time inputs such as current, terminal voltage, and temperature from the ECM, enabling real-time SOC tracking with region-specific noise adaptation to improve convergence and robustness.

Finally, the complete system—comprising the dynamic ECM and the A-SRUKF estimator—is executed in a SIL environment to evaluate the functional accuracy, response stability, and estimation performance under simulated load cycles. This closed-loop validation demonstrates the estimator's ability to adapt to different SOC regions and ensure reliable SOC prediction for battery management systems (BMS) in electric vehicles.

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SYMBOLS & ABBREVATIONS

SYMBOL	ABBREVATION
ECM	Equivalent Circuit Model
SOC	State of Charge
SOH	State of Health
BMS	Battery Management System
HPPC	Hybrid Pulse Power Characterization
OCV	Open Circuit Voltage
Li-on	Lithium-ion
EV	Electric vehicle

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

As Electric Vehicles (EVs) become increasingly widespread, the need for smarter and more efficient battery charging methods is critical. Traditional fixed rate charging techniques often fail to account for real-time changes in battery condition, which can lead to overheating, reduced efficiency, and long-term battery degradation. To overcome these limitations, this project focuses on the concept of adaptive charging, where charging parameters are continuously adjusted based on real-time feedback from the battery.

The primary objective of this project is to enhance battery performance, safety, and lifespan by accurately estimating internal parameters and the State of Charge (SoC) under varying operating conditions. By leveraging real-time feedback such as current, terminal voltage, and temperature, the system dynamically identifies ECM parameters and estimates SoC using advanced Kalman filtering techniques. This approach enables more reliable monitoring, reduces the risk of over-discharge or overcharge, and supports efficient battery management without relying on fixed assumptions or static models.

In this project, simulations are conducted using MATLAB/Simulink to model the EV battery system. A lithium-ion battery is used as the reference battery type due to its widespread use in modern EVs. The charging strategy is implemented by monitoring the battery's condition in real time and adjusting the charging parameters accordingly. An Equivalent Circuit Model (ECM) is used to represent battery behaviour and validate the results.

The study provides insights into how advanced estimation techniques can improve battery monitoring accuracy, system reliability, and real-time decision-making in Battery Management Systems (BMS). The outcomes from the experimental simulations form a foundation for future work in smart EV charging systems, potentially integrating real-time data analytics or AI-based control strategies.

1.2 OBJECTIVE

- To develop a simulation model of an EV battery system using MATLAB/Simulink based on an equivalent circuit model(ECM) for accurate representation of battery dynamics

- To implement an adaptive state estimation algorithm that adjusts discharging parameters such as current, and voltage based on real time battery feedback (SOC, temperature, terminal voltage)

- To provide experimental validation through simulation results that support the feasibility and effectiveness of adaptive charging strategies for EV applications.

1.3 NEED AND MOTIVATION

As electric vehicle (EV) adoption increases, efficient and safe battery charging becomes more critical than ever. Traditional fixed-rate charging methods, which apply a constant current and voltage throughout the discharging cycle, are increasingly inadequate. They fail to adapt to real-time battery conditions, leading to issues like overheating, energy inefficiency, and accelerated battery degradation. In contrast, adaptive charging offers a smart solution by dynamically adjusting the charging current and voltage based on key battery parameters such as State of Charge (SOC), temperature, and terminal voltage.

This real-time responsiveness not only enhances safety by preventing overcharging and thermal stress but also improves energy efficiency by optimizing power delivery throughout the charge cycle. Additionally, adaptive charging significantly extends battery life by reducing stress on cells, especially during high and low SOC regions, and allows for faster charging without compromising long-term health. As EV technologies evolve, adaptive charging stands out as a key innovation for ensuring reliable, safe, and sustainable battery performance.

1.4 SCOPE OF THE WORK

This project models a lithium-ion battery used in electric vehicles (EVs) in MATLAB/Simulink and implements adaptive discharging based on real-time data. An Equivalent Circuit Model (ECM) is used to simulate the battery's behavior. The study analyzes improvements in discharging time, battery health, and reliability, laying the foundation for future AI-based smart charging systems.

1.5 ORGANIZATION OF THESIS

The organization of the thesis is as follows

CHAPTER 1- Introduction

provides an overview of the project, including objectives, motivation, scope, and the overall approach adopted for adaptive charging in EVs.

CHAPTER2-LiteratureReview

This chapter presents a comprehensive literature review of existing methods for State of Charge (SOC) estimation, Equivalent Circuit Model (ECM) modeling, and real-time validation techniques. Special emphasis is placed on Kalman filter-based algorithms and OPAL-RT-based testing approaches.

CHAPTER3–Methodology

This chapter details the methodology adopted in the project. It covers EKF-based parameter identification, A-SRUKF-based SOC estimation, and the simulation setup employed for Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) validation.

CHAPTER4–ResultsandDiscussion

This chapter discusses the results obtained from various stages of the project. It analyzes the performance and effectiveness of the proposed methods using both simulation results and real-time testing outcomes.

CHAPTER 5 –Conclusion and Future Work

This chapter summarizes the key findings of the project. It presents final conclusions and outlines potential future directions for research and practical implementation.

CHAPTER 6 –References and Appendix

This chapter lists the references cited throughout the project. It is followed by an appendix section that includes supplementary visuals, graphs, and additional documentation.

CHAPTER 2

LITERATURE SURVEY

- Kalman Filter-based estimation techniques such as the Extended Kalman Filter (EKF) and Adaptive Unscented Kalman Filter (AUKF) are widely used in Battery Management Systems (BMS) due to their real-time adaptability, ability to handle nonlinearity, and robustness to noise (Zhou et al., 2018; Chen and Wang, 2020).

- Battery parameter extraction using EKF from Hybrid Pulse Power Characterization (HPPC) test data has shown superior accuracy for estimating internal resistance R0R_0, and time constants R1C1R_1C_1 and R2C2R_2C_2, compared to traditional static curve-fitting methods (Sun et al., 2017).

- The AUKF has been effectively applied for State of Charge (SoC) estimation by dynamically adapting process and measurement noise covariances (Q and R), thereby improving performance under nonlinear and time-varying battery conditions (Chen and Wang, 2020; Liu et al., 2021).

- Validation of EKF-AUKF frameworks through Hardware-in-the-Loop (HIL) simulations using OPAL-RT has demonstrated strong real-time performance, stability, and accuracy under dynamic driving profiles (Kumar et al., 2019).

- Traditional SoC estimation methods like Coulomb counting suffer from cumulative integration drift and are highly sensitive to sensor noise, making them unreliable for real-world applications (Wang and Lee, 2016).

- The basic Kalman Filter (KF) assumes linearity and Gaussian noise, which limits its applicability in modeling the nonlinear behavior of lithium-ion batteries (Zhang et al., 2015).

- One major limitation of EKF is its reliance on fixed values for Q and R, which can reduce estimation accuracy and system stability under changing conditions (Liu et al., 2018).

- The Adaptive Unscented Kalman Filter (AUKF), by adjusting Q and R based on innovation and residual analysis, significantly enhances estimation accuracy during rapid load and environmental transitions (Chen et al., 2020).

- AUKF has been shown to reduce SoC root-mean-square error (RMSE) by up to 10% compared to EKF and UKF across standard driving cycles, indicating improved performance (Wang et al., 2021).

- More advanced filters like the GMCC-based Strong-Tracking Adaptive Square-Root UKF improve robustness against non-Gaussian noise and outliers, making them suitable for uncertain battery conditions (Li et al., 2022).

- Adaptive learning-based approaches such as the Adaptive Incremental Learning AUKF integrate ambient temperature feedback through online learning for improved estimation in thermally dynamic environments (Sharma et al., 2022).

- Hybrid methods that combine Adaptive Robust UKF with Recursive Least Squares (RLS) have been proposed to enable simultaneous ECM parameter identification and SoC estimation, enhancing adaptability (Nguyen et al., 2021).

- Joint estimation techniques like the Joint Forgetting Factor EKF allow concurrent tracking of SoC and battery parameters by dynamically adjusting a forgetting factor to account for recent battery behavior (Patel and Roy, 2020).

- Real-time simulations using OPAL-RT have confirmed the robustness and adaptability of AUKF under diverse thermal and load conditions, validating its practical viability for EV applications (Singh et al., 2021).

CHAPTER 3

PROPOSED METHODOLOGY

3.1 INTRODUCTION

This project follows a three-stage framework to implement and validate adaptive charging in EVs. It begins with model parameter extraction using EKF, proceeds to SOC estimation using A-SRUKF, and concludes with system-level real-time validation through OPAL-RT-based HIL testing.

3.1.1 Stage 1: ECM Parameter Identification Using EKF

- **Input Data:** The Hybrid Pulse Power Characterization (HPPC) dataset was used, containing time-series measurements of voltage, current, and SOC collected under a range of dynamic loading conditions. This data captures the battery's transient and steady-state responses essential for model identification.
- **Approach:** An Extended Kalman Filter (EKF) algorithm was implemented in MATLAB to estimate the parameters of a second-order Equivalent Circuit Model (ECM). The state-space formulation incorporated the model dynamics, and recursive filtering was applied to extract time-varying values of the resistive and capacitive components.
- **Region-wise Tuning:** SOC was divided into zones, and EKF was tuned separately for each to improve precision.
- **Output:** Lookup tables for R0, R1, R2, C1, C2 as functions of SOC

3.1.2 Stage 2: SOC Estimation using A-SRUKF

- **Method:** The Adaptive Square Root Unscented Kalman Filter (A-SRUKF) is employed for estimating the State of Charge (SOC). It enhances the traditional UKF by adaptively tuning the process noise covariance matrix (Q) and measurement noise covariance matrix (R) using the innovation sequence. This ensures improved filter performance under varying noise conditions.

- **Implementation:** A MATLAB-based implementation takes terminal voltage, current, and temperature as dynamic inputs. The square-root form improves numerical stability, and the adaptive mechanism allows the filter to respond to changes in system behavior across different regions of battery operation.
- **Benefit:** The use of region-specific, adaptively updated Q and R matrices improves the robustness and accuracy of SOC estimation. It enables the filter to better handle sudden load changes and varying measurement uncertainties, leading to more reliable battery monitoring under real-world conditions.

3.1.3 Stage 3: Simulation and Testing in OPAL-RT

- **Model:** A 2-RC ECM was modeled in Simulink, where all parameter values were fed using 2D lookup tables.
- **SOC Estimation Block:** Integrated MATLAB Function block implemented A-SRUKF algorithm.
- **Simulation Setup:** ECM model fed with current profile from OPAL-RT, which returned terminal voltage and estimated SOC in real time.

3.2 WORKFLOW

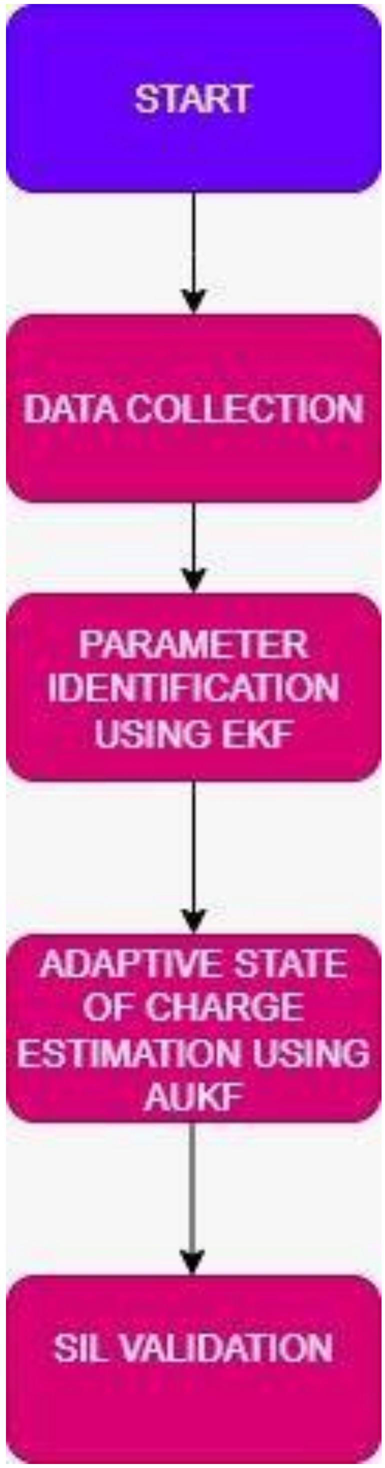

Figure 3.2: Workflow of the Proposed Battery State Estimation Framework using EKF and AUKF and validation

3.3 DATA COLLECTION

3.3.1 REAL TIME EXPERIMENTAL DATA

Region 1		
Time	**Current**	**V_m**
300	0	3.5078
300.5	-100.01	3.3807
301	-100.01	3.3529
301.5	-100.01	3.3306
302	-100.01	3.315
302.5	-100.01	3.304
303	-100.01	3.2959
303.5	-100.01	3.2893
304	-100.01	3.284
304.5	-100.01	3.2795
305	-100.01	3.2756
305.5	-100.01	3.2722
306	-100.01	3.2691
306.5	-100.01	3.2664
307	-100.01	3.2638
307.5	-100.01	3.2614
308	-100.01	3.2593
308.5	-100.01	3.2572
309	-100.01	3.2554
309.5	-100.01	3.2535
310	-100.01	3.2517
310.5	0	3.3253

Table 3.3.1: *Sample HPPC dataset for a SoC region showing time, current, measured voltage (V_m)*

The full dataset used in this study was obtained from a Hybrid Pulse Power Characterization (HPPC) test conducted on a lithium-ion battery. The dataset was segmented into nine distinct regions, each corresponding to a specific State of Charge (SoC) range, covering the full operational window from 100% to 10% SoC in evenly spaced 10% intervals. Each region contains approximately 30 to 50 time-stamped data points, capturing the battery's dynamic response to a constant discharge current pulse (−100.01 A) and subsequent relaxation. For every time step, the dataset records the applied current, measured terminal voltage (V_m), and simulated voltage (V_s). This structured format enables region-wise analysis of internal battery behaviour and supports the estimation of Equivalent Circuit Model (ECM) parameters such as R_0, R_1, R_2, C_1, and C_2 using the Extended Kalman Filter (EKF). Segmentation improves estimation accuracy by accounting for the battery's nonlinear behaviour across different SoC levels.

The HPPC test protocol involved applying a 1C discharge pulse of 100 A for 20 seconds, followed by a 12-minute relaxation period. This process was iteratively carried out across the entire SoC range, with each pulse inducing an SoC drop of approximately 0.5%. After every 10% SoC range (i.e., one complete region), the battery was allowed to rest for one hour to achieve voltage stabilization before initiating the next sequence. This structured cycling approach facilitates the capture of both transient and steady-state behaviour, which is crucial for robust ECM parameter extraction under diverse operating conditions.

3.3.2 PUBLICLY AVAILABLE DATASET

The datasets used in this work include the LA92 driving cycle, which provides the current input profile; the SOC-OCV curve of a Turnigy battery cell measured across temperatures ranging from -20°C to 40°C; and the battery parameter set, which includes internal resistance and second-order RC equivalent circuit components. These datasets enabled realistic simulation and analysis of battery behavior under dynamic load conditions.

3.4 ECM PARAMETER IDENTIFICATION

3.4.1 TWO RC NETWORK

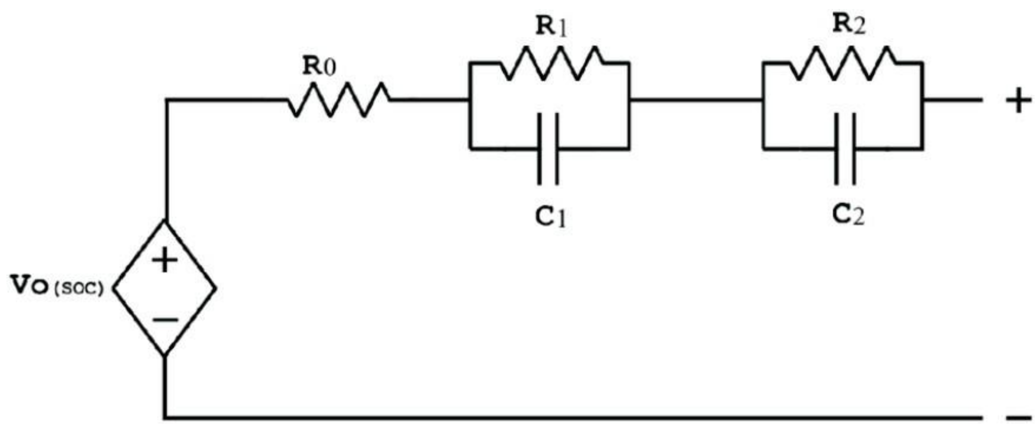

Figure 3.4.1: 2-RC Equivalent Circuit Model (ECM) of a Lithium-ion Battery

The diagram illustrates a 2-RC equivalent circuit model, widely used to represent the dynamic electrical behavior of a lithium-ion battery. The model consists of:

- A voltage source V0(SOC), which depends on the State of Charge (SOC) and represents the open-circuit voltage of the battery.
- A series resistance R0, accounting for the ohmic resistance of the cell (including electrolyte, electrodes, and contacts).
- Two parallel RC networks:
    - R1–C1: models the fast transient response (surface effects and electrolyte reactions).
    - R2–C2: models the slower electrochemical processes, such as diffusion dynamics within the electrodes.

The overall terminal voltage is computed by summing the voltage drops across R0, the two RC branches, and the open-circuit voltage. This structure enables accurate modeling of both steady-state and dynamic responses during battery charging and discharging operations, making it suitable for real-time simulation and state estimation algorithms like EKF and A-SRUKF.

3.4.2 DESCRIPTION

To accurately simulate battery behavior under dynamic load conditions, it is essential to identify the internal parameters of the battery model. The Equivalent Circuit Model (ECM) offers a balance between simplicity and fidelity, making it widely adopted for real-time battery applications. In this project, a 2-RC ECM is used, consisting of an open-circuit voltage source V0(SOC), an ohmic resistance R0, and two RC parallel networks (R1, C1) and (R2, C2), which capture transient and diffusion-related dynamics, respectively.

The internal parameters of the ECM are not fixed; they vary with the battery's State of Charge (SOC), temperature, and aging. Therefore, region-wise estimation is carried out using data from Hybrid Pulse Power Characterization (HPPC) tests. The Extended Kalman Filter (EKF) is employed as an online parameter identification tool, allowing dynamic estimation of R0, R1, R2, C1, and C2 from current and voltage measurements.

3.4.3 PROCEDURE

1. Data Segmentation and Preprocessing

- The HPPC dataset was divided into 9 State of Charge (SoC) regions, each saved in a separate Excel file (e.g., `discharge_pts_region1.xlsx`).
- For each region, time, current, and terminal voltage data were extracted.
- The open-circuit voltage (OCV) for each region was taken from the first voltage value, representing the rest condition.

2. Model Structure

- The ECM used is a second-order RC model, including R0 (ohmic resistance), two RC branches (R1, C1 and R2, C2), and a variable OCV source dependent on SOC.
- The state vector includes: voltage across RC1, voltage across RC2, and the five parameters: R0, R1, C1, R2, and C2.

3. EKF Setup

- The EKF operates on a 7-dimensional state vector and includes process noise (Q) and measurement noise (R) matrices.

- The matrices are adjusted based on the region number to reflect expected variability in model parameters and noise characteristics.
- A measurement noise function is used to scale R dynamically for each region.

4. EKF Estimation Loop (Per Time Step)

For each time step k, the Extended Kalman Filter (EKF) executes the following steps to estimate the internal states of the battery model:

a. State Prediction

This step predicts the internal states (voltages across RC elements and SoC) based on the previous state and current input (current drawn):

$$V\_RC1[k \mid k-1] = \exp(-\Delta t / (R1 * C1)) * V\_RC1[k-1] + (1 - \exp(-\Delta t / (R1 * C1))) * R1 * I[k-1]$$

$$V\_RC2[k \mid k-1] = \exp(-\Delta t / (R2 * C2)) * V\_RC2[k-1] + (1 - \exp(-\Delta t / (R2 * C2))) * R2 * I[k-1]$$

$$SoC[k \mid k-1] = SoC[k-1] - (\eta * \Delta t / Q) * I[k-1]$$

- These equations model the dynamic behaviour of the equivalent circuit elements.
- The voltages across the RC elements (V_RC1 and V_RC2) are updated using their respective time constants.
- The SoC is updated using Coulomb counting, accounting for battery efficiency η and capacity Q.

b. Covariance Prediction

This step predicts the new error covariance matrix to reflect the uncertainty in the predicted state:

$$P[k|k-1] = F[k] * P[k-1] * F[k]^T + Q[k]$$

- F[k] is the Jacobian matrix of the nonlinear state transition function, representing how small changes in the previous state affect the current state.
- Q[k] is the process noise covariance matrix, accounting for modelling errors and system noise.

- This step reflects how the uncertainty in the state propagates over time.

c. Measurement Prediction

The terminal voltage is predicted based on the current estimated states:

V_pred[k] = OCV(SoC[k|k-1]) - V_RC1[k|k-1] - V_RC2[k|k-1] - R0 * I[k]

- This equation estimates the output voltage of the battery using the open-circuit voltage (OCV), RC voltage drops, and ohmic drop.
- The OCV is a nonlinear function of SoC, often obtained from experimental data or a lookup table.
- This is the measurement that will be compared with the actual voltage reading.

d. Innovation Calculation

This step computes the innovation or residual, which is the difference between the actual measured voltage and the predicted voltage:

y[k] = V_measured[k] - V_pred[k]

- This residual indicates how much the model’s prediction deviates from the actual measurement.
- A large innovation suggests a poor prediction and triggers a larger update in the next step.

e. Kalman Gain and State Update

This is the correction step where the predicted state is updated using the innovation:

Kalman Gain:

K[k] = P[k|k-1] * H[k]^T * inv(H[k] * P[k|k-1] * H[k]^T + R[k])

State Update:

x[k] = x[k|k-1] + K[k] * y[k]

Covariance Update:

P[k] = (I - K[k] * H[k]) * P[k|k-1]

- The Kalman gain K[k] determines how much the innovation should influence the new state estimate.
- H[k] is the Jacobian matrix of the measurement function.

- R[k] is the measurement noise covariance, reflecting sensor noise.
- The state vector x[k] is corrected based on the observed discrepancy.
- The error covariance P[k] is updated to reflect the reduced uncertainty after incorporating the measurement.

5. Result Storage

- For each region, the final estimated values of R0, R1, R2, C1, and C2 are stored.
- Time constants $\tau 1 = R1 \times C1$ and $\tau 2 = R2 \times C2$ are computed.
- RMSE is calculated between measured voltage and simulated voltage for validation.

3.4.4 FLOWCHART

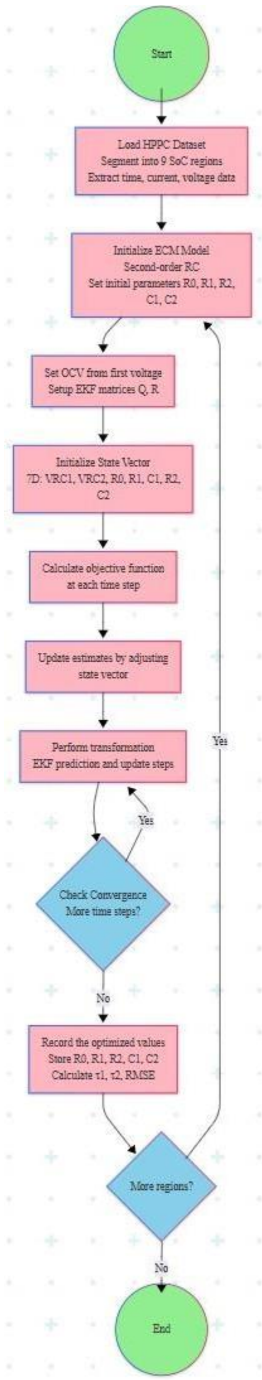

Figure 3.4 Workflow of EKF-Based Parameter Identification for Battery ECM using HPPC Dataset

3.5 Improved Adaptive Square Root Unscented Kalman Filter (ASRUKF) for Lithium-ion Battery SoC Estimation

In this study, we implement an advanced variant of the Unscented Kalman Filter, namely the Adaptive Square Root Unscented Kalman Filter (ASRUKF), to estimate the State of Charge (SoC) of a lithium-ion battery in real time. The algorithm integrates physical modeling (via a second-order Equivalent Circuit Model), statistical estimation, and region-based adaptive noise tuning, while ensuring numerical stability through square-root filtering techniques.

1. Battery Modeling Using a 2-RC Equivalent Circuit Model

The battery is modeled using a 2-RC Equivalent Circuit Model (ECM), which effectively captures both fast and slow dynamic responses of the cell.

The terminal voltage (Vt) is modeled as:

$$Vt = OCV(SOC) - R0 \cdot I - V1 - V2$$

Where:

- OCV(SOC): Open Circuit Voltage as a function of State of Charge (SOC), fitted using an 8th-order polynomial.
- R0: Ohmic resistance.
- V1, V2: Voltage drops across the first and second RC branches, respectively.
- I: Current (positive for discharge).
- R1, R2, C1, C2: Resistances and capacitances of the RC branches.

The model is temperature- and SoC-dependent, with parameters R0, R1, R2, C1, C2 extracted from experimental HPPC data and interpolated using MATLAB's scattered Interpolant.

2. Initial Setup and State Representation

The state vector is defined as:
$\mathbf{x} = [\mathbf{SOC}, \mathbf{V_1}, \mathbf{V_2}]^t \in \mathbb{R}^3$

The initial state estimate is set to **SOC = 0.2**, with **$V_1 = V_2 = 0$**. The state covariance matrix **P** is initialized with large uncertainty in the SoC state to allow aggressive early adaptation. Its **Cholesky decomposition (S)** is used for square-root filtering.

3. Adaptive Region-Based Noise Covariance Tuning

To ensure robust performance across the SoC range, the full range [0, 1] is divided into 9 non-overlapping regions, such as [1.0–0.9], [0.9–0.8], ..., [0.1–0.0]. Each region is associated with its own process noise covariance $Q \in \mathbb{R}^{3x3}$ and measurement noise variance R, obtained via Bayesian optimization.

When the SoC estimate transitions into a new region, and a minimum number of samples have been collected in the current region (≥ 10), the algorithm updates the noise parameters by recomputing their square roots via Cholesky decomposition.

This adaptive strategy allows the filter to remain aggressively tuned in dynamic SoC regions, and more conservative in stable areas—balancing responsiveness and stability.

4. Discretization of System Dynamics

The ECM equations are discretized using the sampling time $\Delta t = 30$ seconds. Time constants for the RC branches are computed as:
$\tau_1 = R_1 \times C_1$
$\tau_2 = R_2 \times C_2$

The discrete-time state transition model is constructed as:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \exp(-\Delta t / \tau_1) & 0 \\ 0 & 0 & \exp(-\Delta t / \tau_2) \end{bmatrix}$$

$$B = \begin{bmatrix} -\eta \times \Delta t / Q_n \\ R_1 \times (1 - \exp(-\Delta t / \tau_1)) \\ R_2 \times (1 - \exp(-\Delta t / \tau_2)) \end{bmatrix}$$

This structure ensures the model remains valid even under fast transients and varying temperatures.

5. Square Root UKF Configuration

The Unscented Kalman Filter (UKF) avoids linearization by propagating a set of deterministically chosen sigma points through the nonlinear system.

Filter parameters used:

- $\alpha = 1$: controls the spread of sigma points
- $\beta = 2$: incorporates prior knowledge about Gaussian distributions
- $\kappa = 0$: secondary scaling parameter

The sigma points χ are generated as:
$\chi_0 = \hat{x}$
$\chi_i = \hat{x} \pm \sqrt{(n + \lambda)} \times S_i$, for i = 1,...,n

Each sigma point is propagated through the system dynamics:
$x_{k+1} = f(x_k, u_k)$
and then used to compute the predicted state mean and covariance.

To ensure numerical stability and computational efficiency, the Square Root UKF (SR-UKF) variant is used, where the covariance updates are handled using Cholesky-based rank-1 updates (cholupdate) and QR decomposition.

6. Measurement Update Step

The predicted measurement z_k is derived for each sigma point using the terminal voltage equation. The measurement mean and covariance are computed, and the Kalman gain is calculated as:

$K_k = P_xz \times (Szz \times Szz^t)^{-1}$

Where:

- P_xz is the cross-covariance between state and measurement
- Szz is the square root of the measurement covariance
- The state estimate is updated as:
  $\hat{x}_k = \hat{x}_k^- + K_k \times (y_k - \hat{z}_k)$

The square root covariance is updated using a Joseph-form update as a fallback if cholupdate fails (e.g., due to negative-definiteness). Regularization and SVD recovery mechanisms are employed to ensure positive-definite covariance updates.

3.5.2 TUNING PARAMETERS

The process noise covariance matrix Q and measurement noise covariance R for each SOC region were obtained using Bayesian optimization. This method allowed for region-wise adaptive tuning, enhancing the robustness and accuracy of the SOC estimation process across varying charge levels.

Region	**Q (3×3 Diagonal Elements) – [Q_1, Q_2, Q_3]**	**R**
1	**[1.30×10^{-10}, 2.88×10^{-7}, 8.96×10^{-8}]**	**6.13×10^{-3}**
2	**[1.21×10^{-10}, 3.47×10^{-11}, 6.12×10^{-12}]**	**1.08×10^{-1}**
3	**[4.98×10^{-9}, 3.20×10^{-5}, 4.43×10^{-11}]**	**1.34×10^{-4}**
4	**[4.05×10^{-11}, 2.04×10^{-8}, 2.11×10^{-10}]**	**1.38×10^{-1}**
5	**[2.58×10^{-10}, 1.24×10^{-4}, 4.26×10^{-9}]**	**1.33×10^{-2}**
6	**[6.58×10^{-11}, 4.58×10^{-8}, 4.32×10^{-9}]**	**1.22×10^{0}**
7	**[3.60×10^{-10}, 2.20×10^{-12}, 1.40×10^{-4}]**	**4.19×10^{0}**
8	**[1.17×10^{-12}, 1.80×10^{-7}, 8.60×10^{-6}]**	**4.47×10^{-2}**
9	**[8.79×10^{-9}, 4.62×10^{-3}, 1.26×10^{-3}]**	**3.04×10^{-1}**
10	**[8.42×10^{-7}, 3.88×10^{-5}, 1.62×10^{-9}]**	**9.36×10^{-1}**

Table 3.5.1: Process noise covariance matrix Q and Measurement noise covariance R for each SOC region

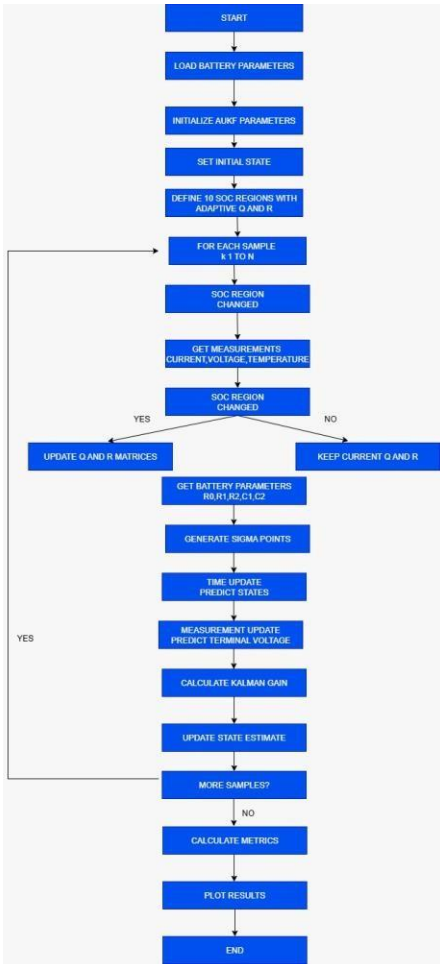

Figure 3.5 Adaptive State of Charge Estimation Using AUKF with Region-Based Q and R Tuning

3.6 SIL VALIDATION

This section outlines the methodology adopted for implementing and validating a battery State of Charge (SOC) estimation algorithm using a Software-in-the-Loop (SIL) simulation framework. The SOC estimation is based on the Adaptive Square Root Unscented Kalman Filter (A-SRUKF), and is tested in conjunction with a 2-RC Equivalent Circuit Model (ECM) of a lithium-ion battery using Simulink. SIL simulation provides an efficient means of testing estimation algorithms in a controlled, purely software-based environment before real-time deployment.

3.6.1. Development of 2-RC Equivalent Circuit Model (ECM)

A second-order ECM of a lithium-ion battery was created in Simulink to represent the electrical behavior of the cell. The model includes:

- Open Circuit Voltage (OCV)–SOC mapping via a 1-D lookup table based on experimental data.
- Series resistance (Rs) representing internal ohmic losses.
- Two RC branches (R1-C1 and R2-C2) capturing the short-term and long-term dynamic response of the battery voltage under load.
- Temperature and SOC-dependent parameters: The resistance and capacitance values vary with both SOC and cell temperature and are modeled using 2-D lookup tables derived from datasheets or test data.

The ECM serves as the virtual battery plant whose output terminal voltage is influenced by input current and internal states.

3.6.2. Application of Ampton Constant Current Discharge Profile

To simulate battery operation, a constant current discharge profile (Ampton profile) was used. This profile involves applying a fixed discharge current continuously over time until the battery reaches its cutoff voltage.

The constant load profile simplifies initial testing and helps evaluate the estimator's ability to track SOC linearly as the battery discharges. This type of test provides a controlled scenario that is ideal for verifying the estimator's stability and convergence characteristics in a predictable environment.

The discharge current is fed into the ECM, which responds by producing terminal voltage data that dynamically changes with SOC and internal voltage drops.

3.6.3. Implementation of the A-SRUKF-Based SOC Estimator

A MATLAB Function block in Simulink was developed to implement the **Adaptive Square Root Unscented Kalman Filter** algorithm. Key features of this estimator include:

- **Input parameters**: Battery current, terminal voltage, cell temperature, and elapsed time.
- **Nonlinear state estimation**: The algorithm models SOC and internal voltage states based on the nonlinear equations derived from the ECM.
- **Adaptive mechanism**: The estimator adaptively updates the process and measurement noise covariance matrices based on innovation statistics, ensuring better performance in the presence of model mismatches or measurement noise.
- **Numerical stability**: Square-root formulation of the UKF ensures more stable matrix operations during runtime compared to the standard UKF.

The estimator runs at every simulation time step, continuously updating its SOC prediction as new voltage and current data becomes available.

3.6.4. SIL Integration and Closed-Loop Simulation

The ECM and A-SRUKF estimator are integrated into a single Simulink model for SIL simulation. In this setup:

- The **ECM acts as the plant**, generating the true voltage response of the battery.
- The **A-SRUKF block acts as the controller/observer**, estimating SOC in real-time based on the current-voltage inputs from the ECM.
- All computation is done in **software only**, simulating the actual estimator code that will later be used in real-time hardware-in-the-loop (HIL) setups.

This closed-loop SIL simulation environment provides a robust platform for evaluating algorithm accuracy, robustness, and convergence behavior.

3.6.5. Output Monitoring and Performance Evaluation

The following outputs were continuously monitored and logged:

- **Estimated SOC** from the A-SRUKF.
- **True terminal voltage** from the ECM.
- **Estimated terminal voltage** from the filter.
- **Estimation error**, i.e., the difference between true and estimated voltages.

These outputs were analyzed using Simulink scopes and logged to the MATLAB workspace for post-simulation evaluation. Performance metrics such as estimation error, convergence time, and response to disturbances (like sudden SOC jumps or noise) were studied.

3.6.6. Simulation Configuration

The simulation was carried out in **MATLAB R2018a** with a fixed-step solver. Parameters used in the ECM and estimator (like sampling time, noise levels, and filter tuning constants) were selected based on experimental constraints and HIL compatibility. The use of constant current discharge allowed for clear validation of estimator linearity and stability.

3.6.7 WORKFLOW

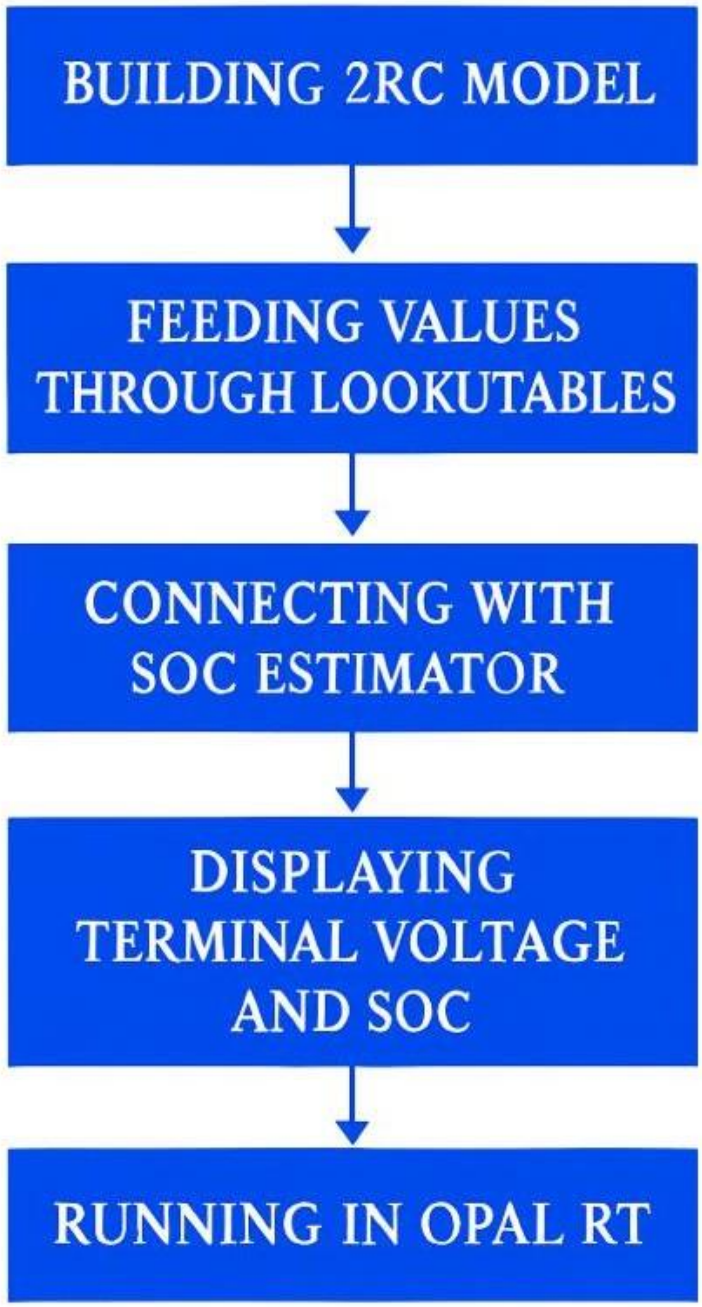

FIG 3.5.1 WORKFLOW SUMMARY OF SIL VALIDATION

BATTERY SPECIFICATIONS

ITEMS	SPECIFICATIONS
Charge voltage	3.65V
Nominal voltage	3.2V
Rated capacity	100 Ah(discharge at 0.5c to voltage of 2.5V)
Rated charge- discharge energy	300Wh
Standard charge – discharge power	150W
Max charge power	300W
Max discharge power	Under conditions of 25 deg c discharging at 600W for 30s at 100% soc)
Discharge cutoff voltage	2.5V
Operating Temperature and humidity	Charging:0-45 deg c,65% Discharging:-20 -60 deg c,65%
Recommended storage temperature	15 -35 deg c
Cell weight	Approx. 2.25kg
Cell dimension	Thick:40.2mm Width:130.2mm Length:220.0mm

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 EVALUATING ECM PARAMETERS

4.1.1 EKF-BASED PARAMETER ESTIMATION OF THE 2-RC MODEL

REGION	R_0 (Ω)	R_1 (Ω)	C_1 (F)	R_2 (Ω)	C_2 (F)	OCV
1	0.0008	0.0011	30506.1	0.0005	34084.4	3.5078
2	0.0012	0.0012	31132.8	0.0004	38227.7	3.3253
3	0.0012	0.0012	28978.6	0.0005	34895.8	3.3261
4	0.0012	0.0012	28077.2	0.0005	34886.6	3.3263
5	0.0012	0.0012	30410.7	0.0004	35707	3.2998
6	0.0012	0.0012	32308.6	0.0005	34720.7	3.2882
7	0.0014	0.0014	28028	0.0005	33459.9	3.2863
8	0.0014	0.0014	29204.8	0.0005	35271.3	3.2845
9	0.0014	0.0014	31794.4	0.0006	35678.5	3.2695

Table 4.1.1: RC parameters for all 9 regions lookup table

4.1.2 ESTIMATION RESULTS AND INTERPRETATION

Ohmic Resistance (R0)

- Increased from 0.0008 Ω (high SoC) to 0.0014 Ω (low SoC)
- Matches known lithium-ion behaviour: increased resistance at low SoC due to reduced ion availability.

Polarization Resistances (R1, R2)

- R1 remained between 0.0011 and 0.0014 Ω
- R2 remained between 0.0004 and 0.0006 Ω
- Indicates consistent impedance behaviour, independent of SoC.

Capacitances (C1, C2)

- Values remained stable across SoC (typically 28000–38000 F)
- Represent dynamic charge storage; not literal physical capacitors.

Table 1 presents the EKF-derived parameter estimates across all SoC intervals. The ohmic resistance R0 increased from 0.0008 ohms at high SoC (80–90%) to 0.0014 ohms at low SoC (10–20%). This trend matches typical lithium-ion behaviour, where internal resistance rises as SoC decreases due to reduced ion availability. Parameter limits (e.g., R0 >= 0.0008 ohms) were enforced to maintain physical consistency.

In comparison, the polarization resistances R1 and R2 remained stable across SoC levels. R1 typically ranged from 0.0011 to 0.0014 ohms, while R2 ranged from 0.0004 to 0.0006 ohms. This indicates that the double-layer and diffusion effects modelled by these parameters are relatively unaffected by SoC changes under test conditions.

Similarly, the estimated capacitances C1 and C2 (constrained between 20,000 and 50,000 farads) showed minimal variation. These large values represent dynamic charge storage behaviour and voltage smoothing and are typical for equivalent circuit models of lithium-ion batteries.

4.1.3 TIME CONSTANT ANALYSIS

Time constants were computed as:

- $\tau 1 = R1 \times C1$ (fast dynamics)
- $\tau 2 = R2 \times C2$ (slow relaxation)

To analyze the dynamic performance of the 2-RC ECM, time constants $\tau_1 = R_1 \times C_1$ and $\tau_2 = R_2 \times C_2$ were computed. These constants, reflecting the short- and long-term voltage relaxation behaviors respectively, remain within 25 to 40 seconds across all SoC regions. This time range confirms that the EKF-based model accurately captures both fast transients and slower relaxation phenomena in the voltage profile, as corroborated by voltage tracking performance during simulation.

4.1.4 CONVERGENCE AND ACCURACY

- EKF successfully converged in all nine SoC intervals.
- Filter remained stable due to tuning of Q and R matrices.
- Voltage tracking was accurate with low Root Mean Square Error (RMSE).
- Innovation (residual) signals remained within expected bounds.

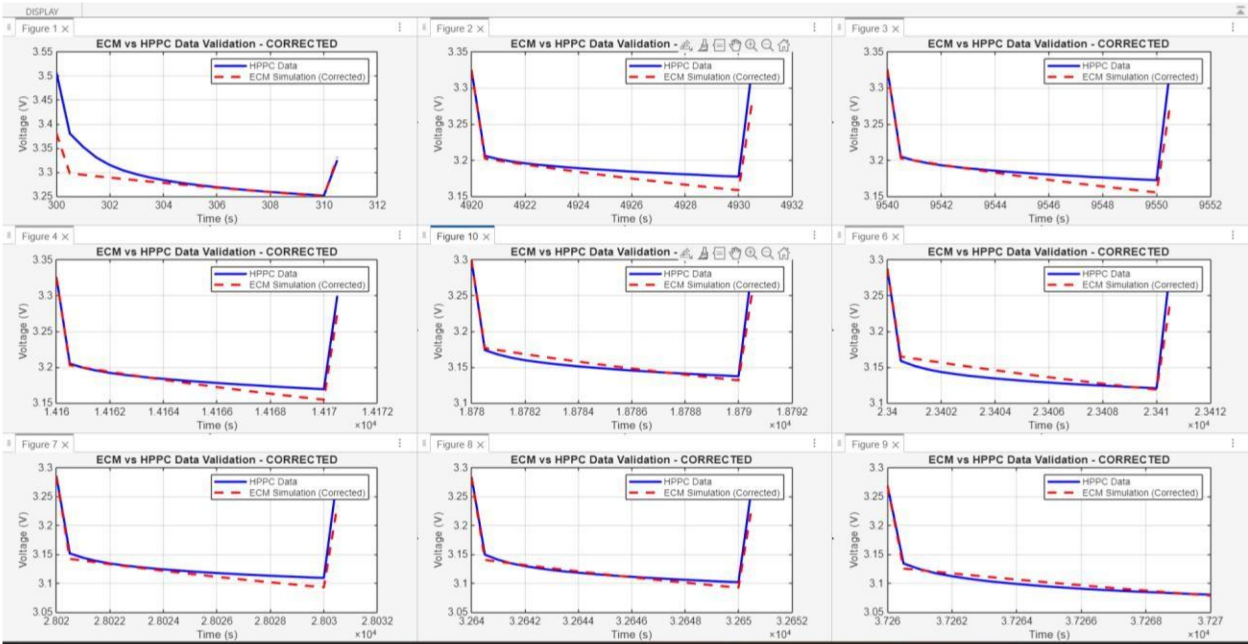

Figure 4.1.2: ECM Voltage Validation Across 9 SoC Regions Using HPPC Data

Comparison between measured terminal voltage from HPPC tests (blue solid line) and the ECM-simulated voltage using region-wise EKF-estimated parameters (red dashed line). Each subplot corresponds to one of the 9 SoC regions used for parameter identification, demonstrating the model's accuracy across a wide range of operating conditions.

Minor deviations observed at the onset of current transients are attributed to the inherent delay in the EKF's response to abrupt changes. These transients, however, quickly converge, indicating that the dynamic RC network responds adequately and reflects the underlying electrochemical processes.

The close alignment of the ECM response to the measured data across all operating regions demonstrates the validity and robustness of the 2-RC model structure and the effectiveness of the EKF algorithm in estimating parameters that accurately capture battery behavior under dynamic load conditions.

4.2 IMPLICATIONS FOR SOC ESTIMATION AND BATTERY MANAGEMENT

The overall consistency of the estimated RC parameters and the strong agreement between model and measurement confirm the suitability of the proposed approach for real-time battery modeling and State-of-Charge (SoC) estimation applications.

The minimal variation in dynamic parameters suggests that a fixed 2-RC model structure may be sufficient for practical SoC observers, with adaptive tuning of only the ohmic resistance R_0 based on SoC or temperature.

These results also support the use of the EKF not only as a state estimator but also as a reliable parameter identifier for model-based diagnostics and control. The capability to capture both immediate voltage drops and slower relaxation dynamics makes the 2-RC ECM a viable choice for integration into Battery Management Systems (BMS) for electric vehicle and grid storage applications

4.2 SOC ESTIMATION BASED ON AUKF

4.2.1 Public Dataset Evaluation

The ASRUKF was first validated on publicly available datasets using ECM parameters already provided. Drive cycles such as UDDS, US06, and FUDS were used.

- Achieved SoC RMSE < 2.01%
- Achieved SoC MAE < 0.65%
- Demonstrated robustness to temperature variation and model noise
- Outperformed fixed-noise UKF and EKF variants

4.2.2 Real-Time Experimental Data

The ASRUKF was also deployed on real-time experimental data collected from a custom-built battery testing setup.

- For real-time data, the ECM parameter R0 ,R1 ,R2 ,C1 ,C2 were estimated using EKF from HPPC discharge tests at various SoC levels and temperatures.
- These parameters were formatted into an interpolation structure for use in the ASRUKF.
- Bayesian-optimized Q and R matrices were used, identical in structure to those used for public dataset evaluation.

Key observations:

- Excellent tracking of SoC under load pulses and regenerative braking

- Robust performance despite sensor noise and fluctuating temperatures
- Smooth adaptation between noise regions with no instability or divergence

4.2.3. Output and Metrics

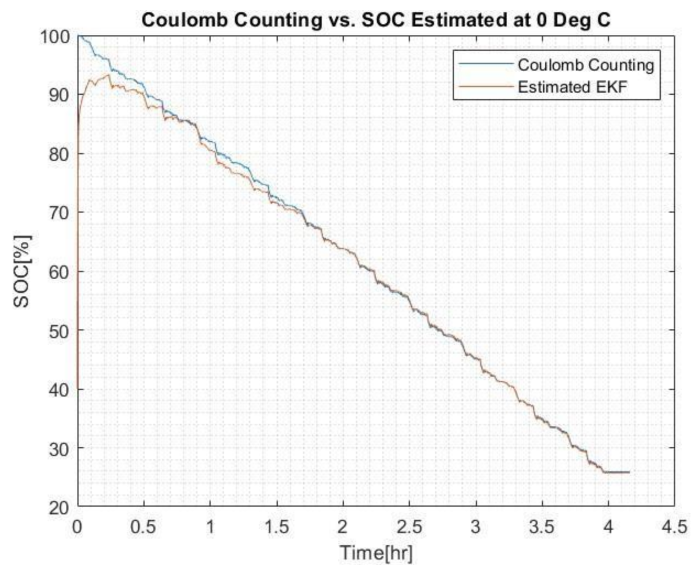

Figure 4.2.1: SOC ESTIMATION FOR PUBLIC DATA SET

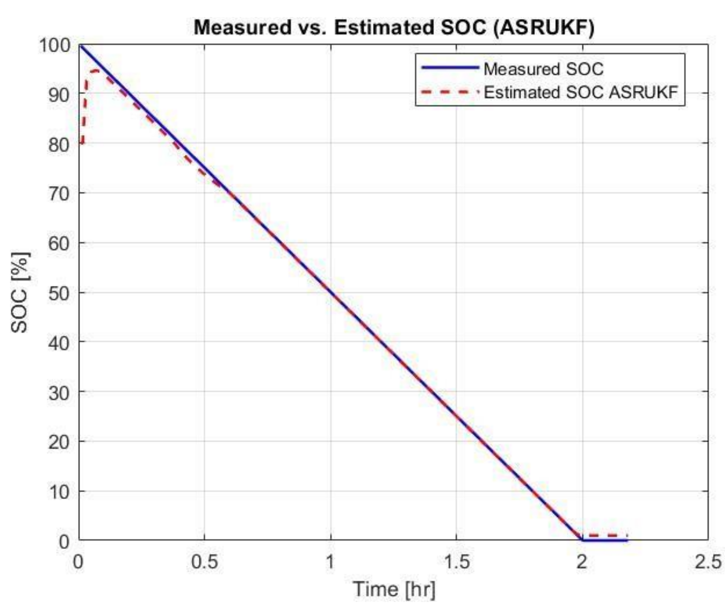

Figure 4.2.2: SOC ESTIMATION FOR REAL TIME DATA SET

The algorithm returns the following key outputs:

- **SOC_Estimated**: Filtered State of Charge (SoC) trajectory over time. This represents the algorithm's estimation of the battery's internal SoC, continuously updated based on voltage, current, and model dynamics.

- **Vt_Estimated**: Estimated terminal voltage of the battery, computed using the second-order equivalent circuit model and the estimated state vector.

- **Vt_Error = Vt_Actual − Vt_Estimated**: Voltage estimation error, which quantifies the difference between the actual measured terminal voltage and the estimated voltage at each time step.

In addition to these outputs, the algorithm also prints the following for performance monitoring:

- **Initial and final SoC values**: Useful for verifying whether the estimation starts and ends in a physically reasonable range.

- **Total number of region transitions**: Indicates how many times the adaptive filter switched between SoC regions during the run, giving insight into how dynamic the operating conditions were.

4.2.4. Advantages and Features

Feature	Description
Region-based adaptation	Dynamically adjusts Q/R based on SoC, improving robustness and tracking
Square-root filtering	Improves numerical stability over conventional UKF
Temperature-aware model	Interpolates ECM parameters from 3D lookup tables based on T and SoC
Fallback handling	Ensures robustness via Joseph-form and SVD recovery
OCV polynomial model	Efficient and accurate OCV computation using 8th-order fit
Real-time feasibility	Implemented efficiently for potential deployment on HIL systems

This algorithm demonstrates high-fidelity tracking of SoC even under abrupt transients and wide temperature/SoC operating ranges. The modular structure also allows seamless extension to include State of Health (SoH) estimation, thermal modeling, or integration with fast-charging controllers.

4.2.5 Performance analysis

1. Terminal Voltage Estimation Errors

Metric	Real time data	Public data
RMSE (V)	0.0150	0.1138
MAE (V)	0.0107	0.0881
Max AE (V)	0.3153	0.3493

2. SOC Estimation Errors

Metric	Real-Time Data	Public Data
RMSE (%)	1.2993	2.0080
MAE (%)	0.7991	0.6545
Max AE (%)	28.4667	19.5834

3. Observations

- VoltageEstimation:
  The real-time dataset yielded significantly lower RMSE and MAE values for terminal voltage estimation compared to the public dataset. This suggests improved voltage prediction accuracy in real-time conditions.
- SOCEstimation:
  The real-time dataset achieved a lower RMSE, indicating better overall SOC estimation accuracy. However, it had a slightly higher Max Absolute Error, suggesting that in rare cases, the estimation deviates more sharply than in the public dataset.
- DataQuantity:
  The real-time dataset involved 14,962 data points, indicating a robust dataset size for performance evaluation.

4. Conclusion

The ASRUKF algorithm demonstrates better voltage estimation accuracy in real-time conditions. For SOC estimation, while real-time performance is generally superior (lower RMSE), it exhibits occasional large deviations (higher Max AE), possibly due to real-world noise or operational variability not captured in the public dataset.

4.3 REAL TIME IMPLEMENTATION OF ECM MODEL IN SIL CONFIGURATION USING REAL TIME CONFIGURATOR

4.3.1 ECM Model Simulation in MATLAB

This subsection presents the simulation results of the Equivalent Circuit Model (ECM) designed in MATLAB Simulink. The ECM incorporates temperature and State of Charge (SOC) dependent parameters for dynamic resistance and capacitance behavior using 2D lookup tables. The SOC and temperature inputs dynamically update internal components such as R0,R1 ,C1 ,R2 and C2allowing a realistic response of the battery terminal voltage under a constant current load of -50 A.

Figure 4.3.2 displays the simulated terminal voltage, which shows a gradual decline from approximately 3.44 V to 3.27 V over a span of 10 seconds. This response is consistent with typical lithium-ion battery discharge behavior and validates the internal configuration of the 2-RC ECM.

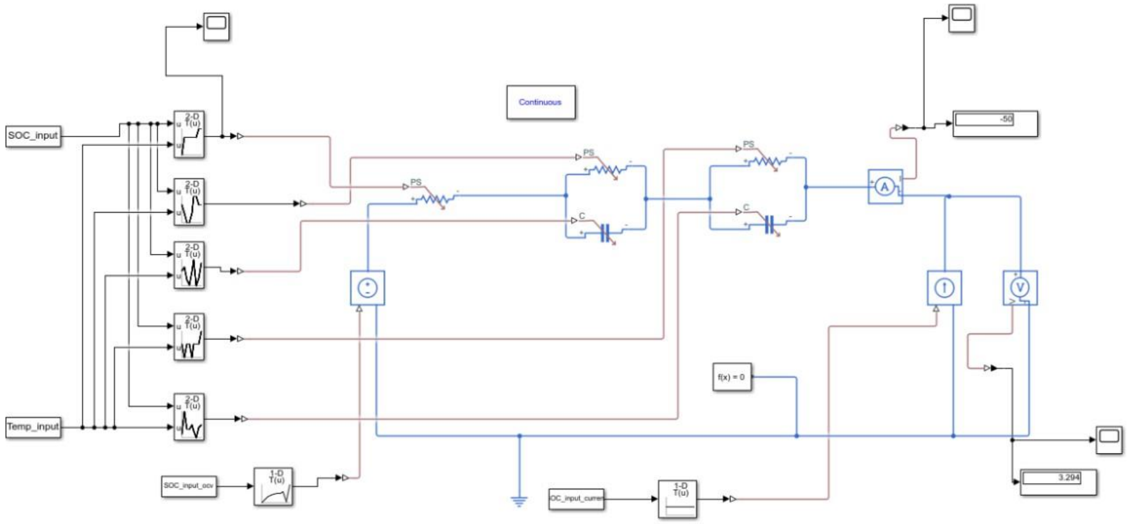

Figure 4.3.1: MATLAB Simulink ECM model

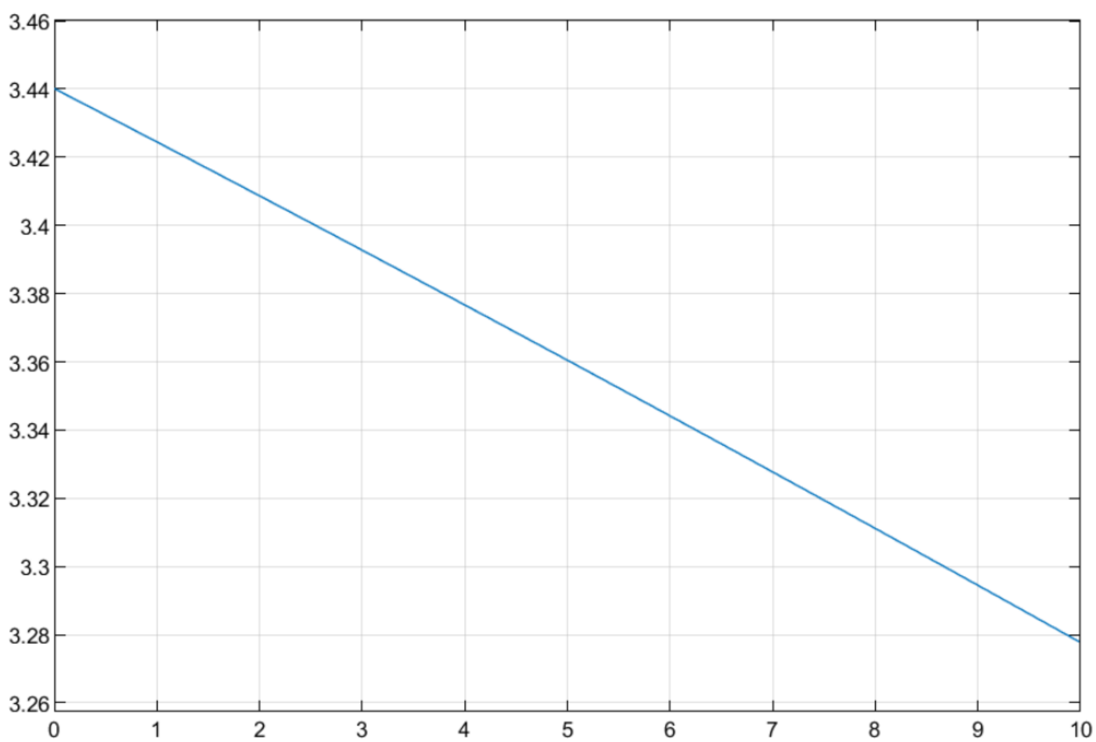

Figure 4.3.2: MATLAB terminal voltage simulation result

4.3.2 SIL Testing in RT-LAB

To validate the ECM model in a real-time environment, Software-In-the-Loop (SIL) testing was conducted using OPAL-RT's RT-LAB platform. The ECM model was deployed and executed in a real-time context to verify the fidelity of its performance.

Figure 4.3.3 illustrates the terminal voltage output obtained from RT-LAB. The voltage curve closely follows the simulated MATLAB result, indicating that the ECM behavior is preserved during SIL implementation. The slight variation, if any, can be attributed to:

- Time discretization in the real-time solver
- Hardware interfacing delays or limitations
- Numerical precision and data handling differences between MATLAB and RT-LAB

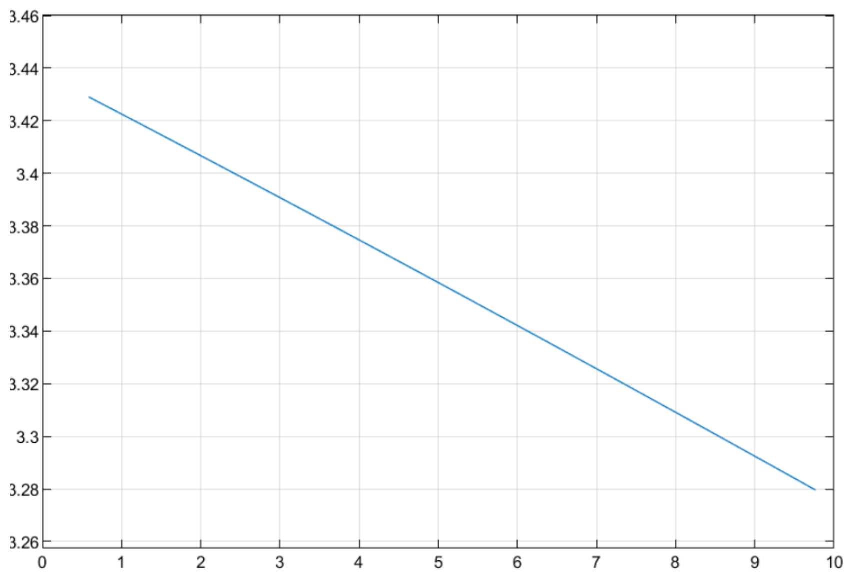

Figure 4.3.3: RT-LAB SIL terminal voltage output

4.3.3 Comparison Between MATLAB and RT-LAB Outputs

METRIC	MATLAB OUTPUT	RT LAB OUTPUT
Initial voltage(v)	~ 3.44	~3.43
Final voltage(v)	~3.27	~3.28
Slope consistency	High	High

The comparison confirms high correlation between the two environments, thereby validating the accuracy and portability of the ECM model. This proves its applicability for real-time simulation and eventual HIL (Hardware-In-the-Loop) implementation.

4.3.4 Observations and Insights

- The 2-RC ECM accurately captures the transient behavior of battery terminal voltage.
- Lookup tables allow the model to adapt based on SOC and temperature, reflecting real-world behavior more reliably.
- Both MATLAB and RT-LAB show consistent voltage drop trends, confirming successful SIL execution.
- Real-time execution did not introduce significant computational delays or instability in the model.

CHAPTER 5

SUMMARY, CONCLUSION AND FUTURE WORK

5.1 SUMMARY

This project focused on developing a comprehensive and adaptive electric vehicle (EV) charging system through accurate battery parameter estimation and robust State of Charge (SoC) tracking. The Equivalent Circuit Model (ECM) parameters — including R0, R1, R2, C1, C2 — were extracted using the Extended Kalman Filter (EKF) for each region of segmented SoC. These values were then used to construct lookup tables for a dynamic Simulink-based 2-RC ECM model. SoC estimation was carried out using an Adaptive Square Root Unscented Kalman Filter (A-SRUKF), which adaptively tuned the process and measurement noise covariances for improved accuracy and robustness. The entire system was validated through both Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) testing on the OPAL-RT platform, demonstrating reliable real-time performance under various load and noise conditions.

5.2 CONCLUSION

The integration of adaptive filtering techniques with real-time simulation and validation tools significantly enhanced the accuracy and reliability of the EV battery management system. Region-wise EKF parameter identification enabled the creation of a highly responsive and representative battery model. Meanwhile, A-SRUKF-based SoC estimation showed strong adaptability and precision, even under dynamic operating conditions. The overall system maintained high tracking fidelity and computational efficiency throughout SIL and HIL simulations, proving the effectiveness of the approach for intelligent EV charging applications. This project demonstrates that adaptive estimation combined with real-time hardware validation forms a scalable and high-performance foundation for modern Battery Management Systems (BMS).

5.3 FUTURE WORK

1. In real-world EV usage, battery performance is heavily influenced by temperature variations and cell aging. Future work can incorporate temperature-dependent resistance and capacity models, and degradation-aware parameters to improve long-term prediction accuracy. This would also enable early fault detection and extend battery life through preventive strategies.

2. The current MATLAB-based estimation framework can be translated into C/C++ for implementation on microcontrollers or DSP platforms within a Battery Management System (BMS). This step would enable live SOC tracking during vehicle operation, ensuring robust energy management and safety.

3. Instead of manually tuning the process and measurement noise covariances (Q and R), machine learning techniques such as reinforcement learning or neural network-based predictors can be used to auto-adjust these matrices based on driving conditions. This would improve adaptability across diverse usage profiles.

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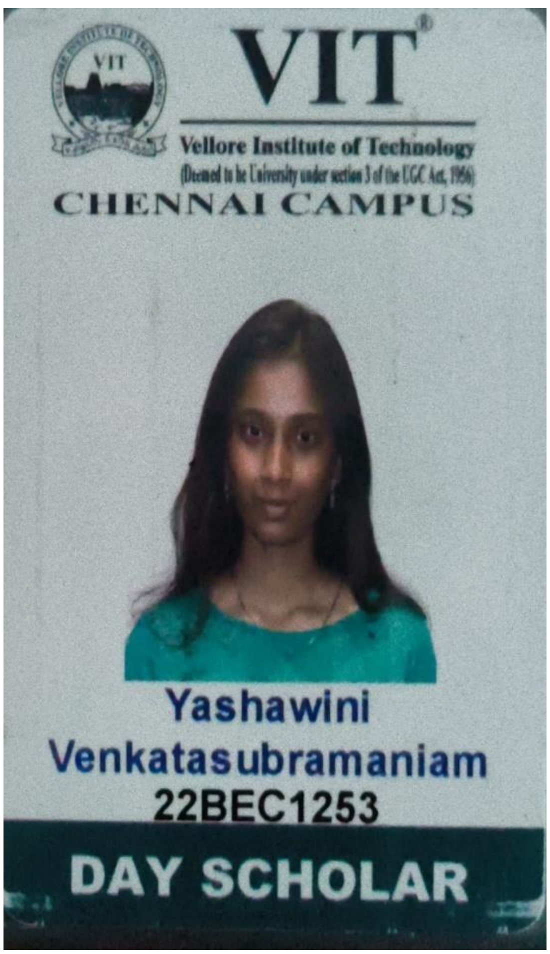
APPENDIX

1. **Scanned copies of student's college ID cards**

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2. Internship working Geotag photos